

XRPL Validator — Daily Engineering Report • 2026-06-16

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 6 commits (xrpl-validator 6)

Engineering changes

- **fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB** (2f0f58c , xrpl-validator) 3.2.0 relocated these out of xrpld/app/*: HashRouter->xrpl/core, LoadFeeTrack->xrpl/server, OrderBookDB->xrpl/ledger.
- **fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)** (b9cf3ce , xrpl-validator) 3.2.0 turned OrderBookDB into a 5-method interface; the concrete OrderBookDBImpl needs a full ServiceRegistry. Single-tx apply crosses offers via the ledger View's book-directory SLEs, not this cache, so a no-op impl is hash-safe.
- **fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)** (13e6ecc , xrpl-validator)
- **feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)** (3ff0144 , xrpl-validator) check_shadow_hash now returns the computed state root (Option<[u8;32]>) instead of bool; the ws_sync FFI-shadow hook feeds it + header fields into lockstep::evaluate_ledger when XRPL_LOCKSTEP_SHADOW=1, recording a process-wide LockstepMeter. Shadow-only, no signing/state impact.
- **feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)** (294e85c , xrpl-validator)
- **fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0** (7e5b1b9 , xrpl-validator) 3.2.0 peers legitimately send bulk frames up to kMaximumMessageSize=megabytes(64) (overlay/Message.h; 26-bit payload header field). The stale 4 MiB cap silently dropped ~60 MB messages from many peers (116 codec errors observed post-upgrade). Not consensus-critical (state syncs from the reference node, not peer bulk) but correct protocol behavior. Tests: oversized boundary now 65 MiB; new test asserts sub-cap accept.

Appendix — commits

Time	Repo	Commit	Subject
00:03	xrpl-validator	2f0f58c	fix(ffi): 3.2.0 header paths for HashRouter/LoadFeeTrack/OrderBookDB
00:05	xrpl-validator	b9cf3ce	fix(ffi): NoopOrderBookDB for 3.2.0 (OrderBookDB now pure-virtual)
00:06	xrpl-validator	13e6ecc	fix(ffi): default HashRouter::Setup (setup_HashRouter moved to xrpld in 3.2.0)
12:13	xrpl-validator	3ff0144	feat(lockstep): env-gated ledger_hash shadow eval at the FFI hook (M1 task 4)
12:13	xrpl-validator	294e85c	feat(lockstep): expose xrpl_lockstep_shadow_match_ratio gauge (M1 task 5)
12:27	xrpl-validator	7e5b1b9	fix(peer): raise codec MAX_MESSAGE_SIZE 4MiB -> 64MiB to match rippled 3.2.0